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Comparative Analysis of Machine Learning Models for Emotion Classification in Song Lyrics

BCS-5A

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This project has been an extensive learning experience, allowing us to explore classical machine learning, hybrid architectures and modern transformer-based LLMs in a real-world classification setting. We are thankful for the opportunity to work on a project that strengthened both our technical skills and our research intuition.

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Abstract

This project evaluates multiple approaches for multi-class text classification on a song-lyrics emotion dataset consisting of four emotions: anger, calm, happy and sad. Three modeling paradigms were compared:

1. classical machine learning models using TF-IDF features
2. hybrid models combining pretrained transformer embeddings with classical learners
3. end-to-end transformer-based LLM classifiers

Classical models achieved the highest performance (~80%), hybrid models performed slightly lower (~79–80%), whereas transformer models achieved the lowest performance (~74–76%). Additional experiments were conducted on the GoEmotions dataset to determine whether the limitation was due to dataset characteristics rather than model architecture. On the new dataset, the performance hierarchy reversed: LLMs became the strongest (93%), classical models achieved ~88% and hybrid models reached ~88%. This demonstrates that the original dataset limited model performance, not the modeling approach.

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1. Introduction

Emotion classification is a core task in natural language processing (NLP), enabling machines to interpret human affect from text. Its applications span music recommendation, user-behavior analysis, digital assistants, social media monitoring and mental-health evaluation. While much research has explored emotion classification in structured text such as reviews or conversational datasets, far less attention has been given to song lyrics, a domain characterized by metaphor, figurative expressions, non-standard grammar and artistic abstraction. These characteristics introduce unique challenges for NLP systems, making song-emotion prediction an interesting and difficult research problem.

This project investigates how different modeling paradigms perform on this type of creative and ambiguous text. In particular, the study evaluates three major families of models:

1. Classical Machine Learning Models (Logistic Regression, Linear SVM, XGBoost)
2. Hybrid Architectures that combine transformer embeddings with classical classifiers
3. Transformer-based Large Language Models (LLMs) fine-tuned end-to-end for emotion prediction

The purpose of this work is not only to implement these methods but also to understand how their performance varies with respect to text style, linguistic noise and dataset quality. Song lyrics provide a rich testing ground to compare these models, as they often challenge the assumptions under which many NLP systems operate.

To deepen the analysis, the models are also evaluated on a second dataset GoEmotions, representing a more structured and cleaner form of text. This dual-dataset approach allows the project to explore how dataset characteristics influence model behavior and to examine whether classical, hybrid, or transformer architectures generalize better across varying text domains.

2. Literature Review

Emotion classification in text has been widely studied across different modeling paradigms, particularly classical machine learning approaches, transformer-based language models and hybrid embedding-based pipelines. This section briefly reviews the most relevant prior work aligned with the scope of this project.

2.1 Classical Machine Learning Approaches

Classical algorithms such as Logistic Regression, Support Vector Machines (SVM) and XGBoost have long served as strong baselines for text classification, especially when paired with vectorization techniques like TF–IDF. Prior studies show that these models perform well when datasets are small, structured, or domain-specific, due to their ability to generalize with limited parameters and strong regularization [1].

However, classical models are limited by their reliance on bag-of-words or n-gram features, which fail to capture deeper semantic meaning, figurative language and contextual dependencies [2]. These limitations are particularly relevant for domains such as song lyrics, where emotions are often metaphorically expressed.

2.2 Transformer-Based Language Models (LLMs)

Transformers such as BERT, RoBERTa, DistilRoBERTa and DeBERTa have redefined NLP by introducing self-attention mechanisms capable of capturing long-range dependencies and nuanced context [3]. When fine-tuned on emotion-annotated corpora, these models typically achieve state-of-the-art results in emotion and sentiment classification.

However, several studies show that transformers may underperform on poetic or creative text due to domain mismatch between pretraining corpora (Wikipedia, books, web text) and the metaphor-heavy nature of lyrics [4]. Their performance also depends heavily on dataset size, annotation quality and fine-tuning stability.

2.3 Hybrid Models

Hybrid pipelines combine the strengths of both paradigms by extracting context-rich embeddings from pretrained LLMs (e.g., Sentence-BERT, DistilRoBERTa) and using them as input to classical classifiers like SVM or XGBoost. Literature shows that hybrid models often outperform end-to-end fine-tuned LLMs on small, imbalanced, or noisy datasets, because the classical classifier reduces overfitting while the embeddings preserve semantic richness [5].

Emotion-specific embedding models, such as RoBERTa-based emotion encoders, have demonstrated notable improvements in tasks where domain-aligned representations are essential.

2.4 Challenges in Emotion Classification of Song Lyrics

Multiple studies highlight unique difficulties associated with lyric-based emotion recognition:

- Figurative language and metaphors reduce the reliability of literal semantic cues, making it harder for models to extract emotion from non-literal expressions [6].
- Emotional ambiguity is common, as lyrics often express overlapping or mixed emotional states rather than a single dominant label.
- Domain mismatch arises because most NLP models are trained on prose or conversational text, not poetic or musical language structures.
- Subjectivity in annotation: human raters frequently disagree on emotional interpretation, leading to noisy and inconsistent labels

These challenges are widely documented in prior NLP research and highlight why lyric-based emotion detection remains a complex task compared to standard text classification domains.

3. Dataset Description

This study uses two datasets to explore emotion classification performance across varying text domains:

- a song lyrics emotion dataset, which serves as the main experimental dataset
- a benchmark conversational emotion dataset, used to assess how dataset quality and linguistic characteristics influence model behavior.

Both datasets differ significantly in structure and language style, making them suitable for comparative analysis.

3.1 Primary Dataset: Song Lyrics Emotion Dataset

The primary dataset used in this project is the “Annanay/aml_song_lyrics_balanced” collection, consisting of approximately 12.2k English song lyric excerpts in total, with around 2.54k samples reserved for the test split. The dataset is evenly distributed across four emotion categories:

- Anger (3,049 samples)
- Calm (3,049 samples)
- Happy (3,049 samples)
- Sad (3,049 samples)

3.1.1 Dataset Structure

Each record in the dataset contains the following fields: lyrics_filename, the name of the source song file; mood, a single categorical emotion label assigned by the dataset author; lyrics, a short excerpt taken from the song; and mood_cats, the corresponding numeric label used for model training. Example entries:

Unnamed	lyrics_filename	mood	lyrics	mood_cats
0	Five_Finger_Death…	anger	"Bulletproof Lyrics[Verse 1]\nYou..	3
1	Karen_O_And….	sad	"Hideaway LyricsHideaway\nWell...	1

The dataset is perfectly class-balanced, limiting class bias during training and evaluation.

3.1.2 Linguistic Characteristics

Song lyrics differ from conventional prose and conversational text due to several unique properties:

- **Figurative and Metaphorical Language:** Lyrics frequently use metaphors, analogies, symbolism and abstract expressions, making literal semantic interpretation difficult.
- **Emotional Ambiguity:** A single lyric line may reflect mixed emotions or subjective interpretations, leading to noisy labels and lower annotation consistency.
- **Non-standard Syntax and Structure:** Lyrics often break grammatical rules, include artistic formatting and rely on rhythm and repetition rather than formal sentence structure.
- **Domain Shift Relative to NLP Pretraining:** Most NLP models, particularly LLMs, are trained on web text, books and conversational data. They do not naturally adapt to poetic, creative, or musical phrasing.

These characteristics make this dataset inherently challenging for computational models.

3.2 Secondary Dataset: GoEmotions (6-Class Subset)

The secondary dataset used in this project is the "dair-ai/emotion" dataset, a refined 6-class subset of Google's GoEmotions project. It contains short, conversational English text sourced from social media platforms. The dataset includes a total of 18,000 samples, with 16,000 allocated for training and 2,000 reserved for testing. The emotion labels are represented across six categories:

- Sadness (4,666 samples)
- Joy (5,362 samples)
- Love (1,304 samples)
- Anger (2,159 samples)
- Fear (1,937 samples)
- Surprise (572 samples)

3.2.1 Dataset Structure

Each record in the dataset contains the following fields: text, a short conversational sentence, label, an integer representing one of the six emotion categories (0 = sadness, 1 = joy, 2 = love, 3 = anger, 4 = fear, 5 = surprise). Example entries:

text	label
i climbed the hill feeling frustrated that id pretty much paced….	3 (anger)
i can t imagine a real life scenario where i would be emotionally connected…	1 (joy)

3.2.2 Linguistic Characteristics

GoEmotions differs from song lyrics in several ways:

a. Sentences are short, direct and conversational
b. Emotional cues are typically explicit rather than figurative
c. Annotations are high-quality and manually validated
d. Language structure closely resembles data used to train modern LLMs
e. Minimal use of metaphors compared to lyrical text

This dataset represents a clean, structured baseline for evaluating emotion classification models.

3.3 Rationale for Using Two Datasets

Using two datasets with contrasting linguistic characteristics allowed the project to conduct a more rigorous and meaningful evaluation of emotion classification models. The lyrics dataset represents creative, metaphorical and highly ambiguous text, whereas GoEmotions consists of short, direct conversational sentences. Evaluating models across these two domains makes it possible to compare how algorithms behave in artistic versus natural language settings and to assess whether the models generalize beyond a single writing style. Additionally, the performance gap observed across the datasets helps isolate whether limitations arise from the models themselves or from the inherent difficulty and noise within the dataset. By benchmarking results on a well-established dataset like GoEmotions, the study ensures stronger research validity and alignment with prior literature, confirming that improvements or failures are meaningful rather than artifacts of dataset quality.

4. Methodology

This project followed a multi-stage methodological framework designed to systematically evaluate three major families of models, Classical Machine Learning models, Hybrid Embedding-based models and Transformer-based Large Language Models (LLMs) on the task of emotion classification. The methodology was designed to ensure a fair, reproducible, and systematic comparison across different modeling strategies while accounting for the unique linguistic characteristics of song lyrics.

4.1 Overview of Experimental Process

The experimental workflow progressed in three major phases.

- First, classical machine learning models were developed to establish baseline behaviour on the lyrics dataset.
- Second, hybrid models combining pretrained transformer embeddings with classical classifiers were introduced to test whether richer semantic representations could improve performance.
- Finally, full transformer-based LLMs were fine-tuned end-to-end on the dataset to explore whether modern deep architectures could capture lyrical emotional cues more effectively.

An additional refinement to the methodology involved introducing a second benchmark dataset (GoEmotions 6-class) after initial experiments indicated that performance on lyrics was consistently lower than expected. This allowed the study to separate model limitations from dataset limitations and to strengthen research insights through cross-dataset validation.

4.2 Data Preprocessing

A uniform preprocessing pipeline was applied across all experiments to ensure consistency. The text was normalized through lowercasing and light punctuation cleaning while intentionally preserving artistic structures such as line breaks and repetitions. Non-textual artifacts (e.g., URLs) were removed. For classical models, text was vectorized using TF-IDF with carefully tuned n-gram settings. For hybrid and transformer models, tokenization was performed using HuggingFace tokenizers with fixed sequence lengths between 96 and 128 tokens, depending on the model architecture. This preprocessing strategy ensured compatibility across models while minimizing loss of semantic or stylistic information that could influence emotion detection.

4.3 Classical Machine Learning Models

Classical models served as the foundation of the analysis and were essential for establishing interpretable baselines. Three families of models were implemented: Logistic Regression,

Linear Support Vector Machines (SVM) and XGBoost, along with a soft-voting ensemble combining all three.

4.3.1 Iterative Improvement of Logistic Regression

Development of Logistic Regression alone required 17 progressively refined iterations.

Early implementations used basic unigram TF-IDF vectors and default hyperparameters. Subsequent iterations introduced systematic refinements including:

- Expansion from unigrams to bi-, tri- and eventually 4-grams
- Adjustments to TF-IDF parameters including *min_df, max_df* and *max_features*
- Experiments involving the removal and reintroduction of stopwords
- Tuning of the regularization strength (C), solver selection and class balancing
- Feature pruning to mitigate overfitting

Each refinement aimed to better capture emotional phrasing patterns typical in lyrics such as repeated motifs, multi-word expressions and stylistic cues.

4.3.2 Additional Classical Models

After Logistic Regression stabilization, Linear SVM and XGBoost were implemented using optimized TF-IDF vectors. The soft-voting ensemble was then constructed to aggregate probabilistic outputs from all three classifiers, providing a more robust baseline for comparison against hybrid and transformer-based approaches.

4.4 Hybrid Models (LLM Embeddings + Classical ML)

The second phase of experimentation introduced hybrid models designed to test whether high-quality transformer embeddings could enhance classical classification performance. Instead of relying on TF-IDF, raw lyrics were transformed into dense vector embeddings prior to classification.

4.4.1 Embedding Generation

Multiple embedding backbones were explored:

- distilroberta-base
- distilbert-base-uncased
- roberta-base
- Emotion-tuned RoBERTa models (*j-hartmann* series)

Embeddings were generated using SentenceTransformers, utilizing mean-pooling layers to obtain sentence-level representations. These embeddings incorporate contextual and semantic information beyond surface-level vocabulary patterns captured by TF-IDF.

4.4.2 Classical Back-End

The embeddings served as input features for the 3 classical models (Logistic Regression, Linear SVM, XGBoost). Multiple iterations (Hybrid1–Hybrid3) varied embedding models, normalization strategies and classifier configurations. This phase tested whether pretrained language representations could compensate for stylistic irregularities and figurative language in song lyrics..

4.5 Transformer-Based LLM Fine-Tuning

The final methodological phase involved fine-tuning full transformer models end-to-end on the lyrics dataset. Five large-scale pretrained LLMs were evaluated:

- BERT-base
- RoBERTa-base
- DistilRoBERTa
- DeBERTa-v3-base
- RoBERTa-large

4.5.1 Training Procedure

A custom training pipeline was implemented to maintain fine-grained control over optimization and evaluation. Key components included:

- HuggingFace Transformers for model loading and tokenization
- Cross-entropy loss with class weights to handle distributional imbalance
- AdamW optimizer with weight decay
- Linear warmup scheduler to stabilize early training
- Gradient accumulation to simulate larger batch sizes
- Gradient clipping to prevent exploding gradients
- Early stopping based on validation macro-F1
- Separate train/validation/test loaders for controlled evaluation

This procedure ensured consistency across models and mitigated instability caused by the small size and artistic nature of the lyrics dataset.

4.6 Cross-Dataset Validation

To investigate whether limited performance stemmed from dataset complexity rather than model capability, all three families of models (classical, hybrid and LLM) were re-evaluated on the GoEmotions (6-class) dataset.

The same preprocessing, embedding and training pipelines were reused, with adjustments only to sequence length and batch size where required.

This cross-dataset methodology enabled validation of:

- Model robustness
- Sensitivity to linguistic domain shifts
- Limitations imposed by figurative text
- Comparative advantages of each model family under varying dataset quality

4.7 Evaluation Metrics

All models were evaluated uniformly using the metrics prescribed by the course guidelines:

- Accuracy
- Precision, Recall and F1-Score
- Exact Match (EM)
- Macro AUC (One-vs-Rest)
- Top-k Accuracy
- Confusion Matrices
- Training and validation loss/accuracy curves (for LLMs)

These metrics were selected to ensure comprehensive performance assessment across both balanced and imbalanced emotion classes.

5. Results

The empirical results obtained from all experiments conducted across the three modeling paradigms: Classical Machine Learning, Hybrid Models and Transformer-based Large Language Models (LLMs) are presented. Results are reported separately for the primary lyrics dataset and the secondary dataset, as each dataset demonstrated distinct behavioral characteristics across models. All metrics specified in the project guidelines (Accuracy, Precision, Recall, F1-Score, AUC, Exact Match (EM), Top-k Accuracy and Confusion Matrices) were computed for each model.

5.1 Results on the Lyrics Emotion Dataset

The lyrics dataset proved to be linguistically challenging due to its figurative style, non-literal expressions and ambiguous emotional phrasing. Across all modeling paradigms, performance remained below typical expectations for emotion classification tasks. The quantitative results obtained are summarized.

5.1.1 Classical Machine Learning Models

Classical models achieved the highest performance on the lyrics dataset among all approaches. After extensive tuning across 17 versions, Logistic Regression, Linear SVM and XGBoost converged toward a stable accuracy ceiling.

Model	Accuracy	Macro F1
Soft Voting Ensemble (LR + SVM + XGB)	0.8049	0.8050
Linear SVM	0.8000	0.7991
Logistic Regression	0.7980	0.7967
XGBoost	0.7865	0.7877

Key Observation: All classical models plateaued tightly around ~80% accuracy, demonstrating strong baseline suitability but also a clear performance ceiling on this dataset.

The Soft Voting Ensemble (LR + SVM + XGB) represents the highest-performing classical approach and establishes the upper-bound benchmark (~80%) on the lyrics dataset. The classifier demonstrates excellent separation for anger and calm, with near-perfect recall. Moderate confusion between happy and sad, which is typical for figurative textual content like lyrics. Strong AUC and Top-k metrics, showing the model understands emotional likelihoods even when the top-1 prediction is incorrect.

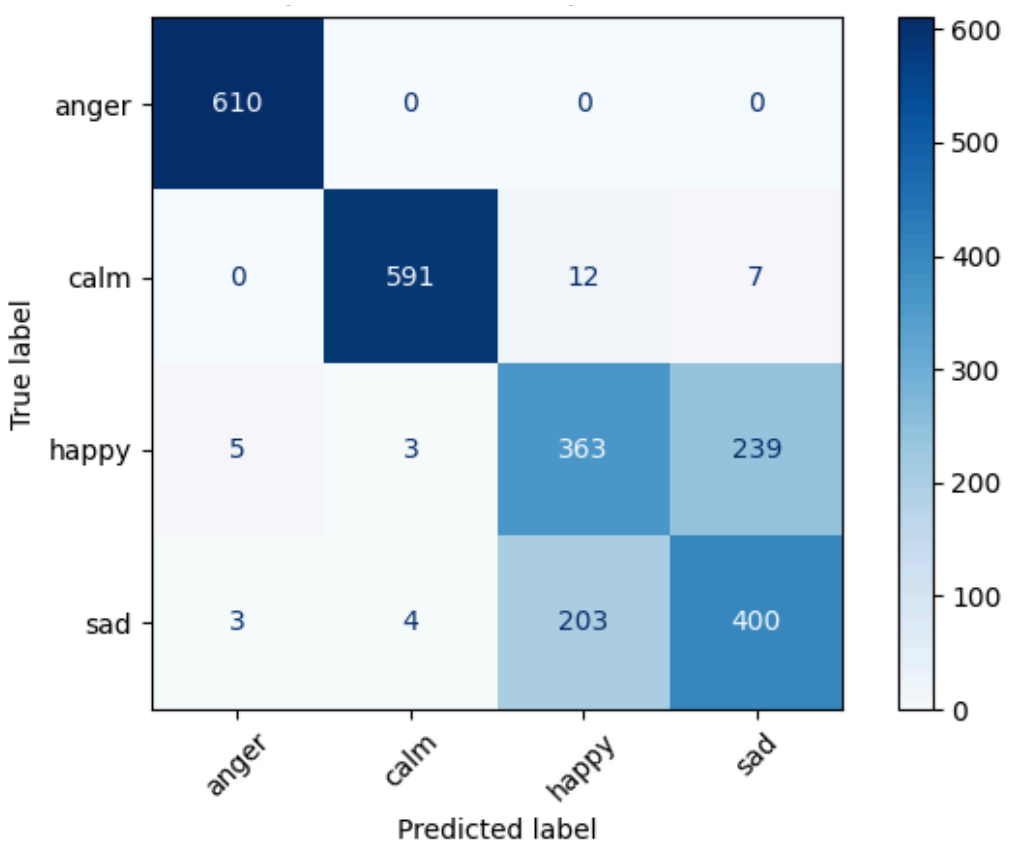
Performance Summary:

Metric	Value
Accuracy	0.8049
Exact Match (EM)	0.8049
Macro F1-Score	0.8050
Macro AUC (OvR)	0.9368
Top-2 Accuracy	0.9807
Top-3 Accuracy	0.9984

Classification Report:

Class	Precision	Recall	F1-Score	Support
anger	0.99	1.00	0.99	610
calm	0.99	0.97	0.98	610
happy	0.63	0.60	0.61	610
sad	0.62	0.66	0.64	610
accuracy			0.80	2440
macro avg	0.81	0.80	0.81	2440
weighted avg	0.81	0.80	0.81	2440

Confusion Matrix:



5.1.2 Hybrid Models

Hybrid models were expected to outperform traditional classical approaches by incorporating contextual embeddings from pretrained transformers. However, on the lyrics dataset, this was not observed.

Embedding Backbone	Classifier	Accuracy	Macro F1
roberta-base	XGBoost	0.7947	0.7944
distilbert-base-uncased	LR/SVM/XGB	0.4635/0.4639/0.7939	0.4553/0.4486/0.7906
distilroberta-base	XGBoost	0.7902	0.7877

Key Observation: Hybrid results remained consistently just below classical models. Although embeddings captured richer contextual semantics, they did not translate into improved separability for lyrical emotional content.

The Hybrid XGBoost model (roberta-base embeddings) is the strongest hybrid approach, reaching ~79% accuracy which is slightly below the classical upper bound. While contextual embeddings improved representation quality, they did not fully resolve the emotional

ambiguity present in lyrical text. The model performs strongly on anger and calm, but still struggles to clearly separate happy and sad indicating that semantic richness alone is insufficient for metaphor-heavy inputs. Its high AUC and excellent Top-k scores show that the model often ranks the correct emotion highly, even when the top prediction is incorrect.

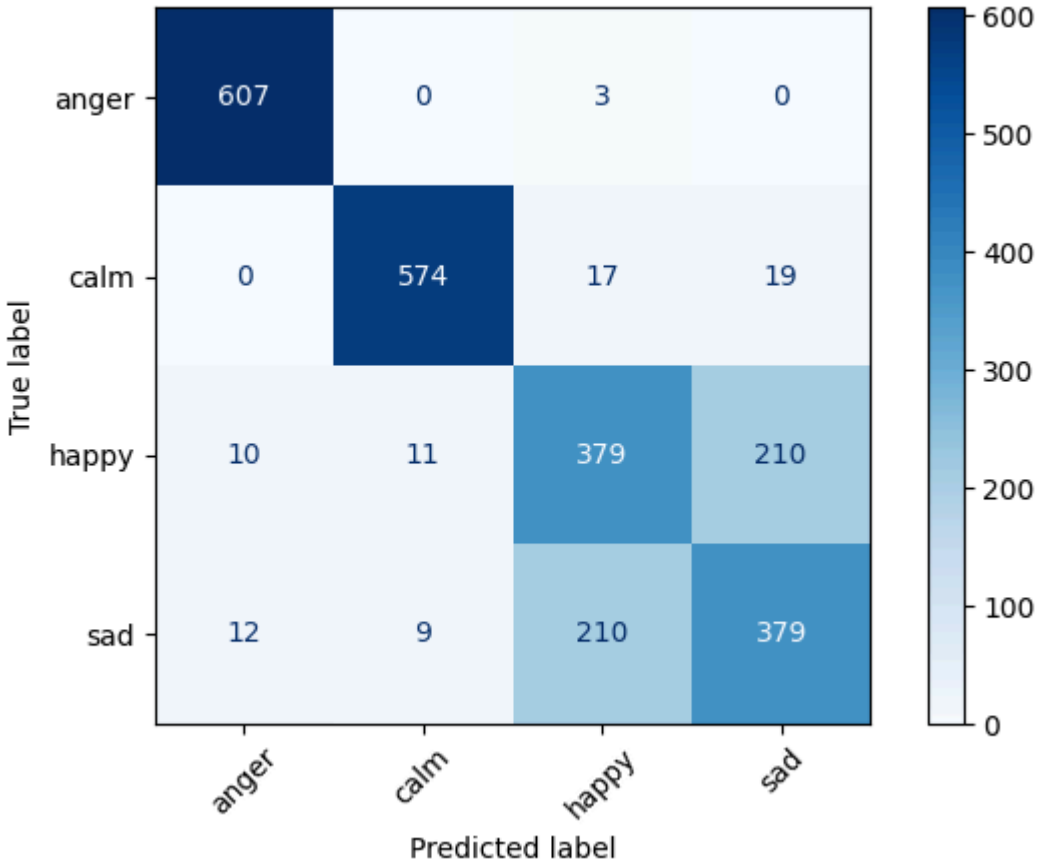
Performance Summary:

Metric	**Value**
Accuracy	0.7947
Exact Match (EM)	0.7947
Macro F1-Score	0.7944
Macro AUC (OvR)	0.9391
Top-2 Accuracy	0.9488
Top-3 Accuracy	0.9975

Classification Report:

Class	**Precision**	**Recall**	**F1-Score**	**Support**
anger	0.97	1.00	0.98	610
calm	0.97	0.94	0.95	610
happy	0.62	0.62	0.62	610
sad	0.62	0.62	0.62	610
accuracy			0.79	2440
macro avg	0.79	0.79	0.79	2440
weighted avg	0.79	0.79	0.79	2440

Confusion Matrix:



5.1.3 Transformer-Based LLM Fine-Tuning

Five transformer models were fine-tuned end-to-end on the lyrics dataset using a controlled training loop with class weighting, gradient accumulation, learning rate warmup and early stopping.

Model	Accuracy	Marco F1
BERT-base	0.7574	0.7479
RoBERTa-large	0.7500	0.7374
DeBERTa-v3-base	0.7426	0.7299
RoBERTa-base	0.7332	0.7202
DistilRoBERTa	0.5988	0.5871

Key Observation: Despite their complexity, transformer-based LLMs achieved the lowest performance among the three paradigms on the lyrics dataset. This highlights a strong domain mismatch between the training corpora of LLMs and the poetic, metaphor-heavy structure of song lyrics.

BERT-base emerged as the best-performing transformer on the lyrics dataset, reaching roughly 75% accuracy. It handles clear emotions like anger and calm effectively but also struggles with the subtle distinction between happy and sad. Despite its lower top-1 accuracy compared to classical models, its strong AUC and high Top-k scores indicate that the correct label often appears among its top-ranked predictions, even when not selected as the final output.

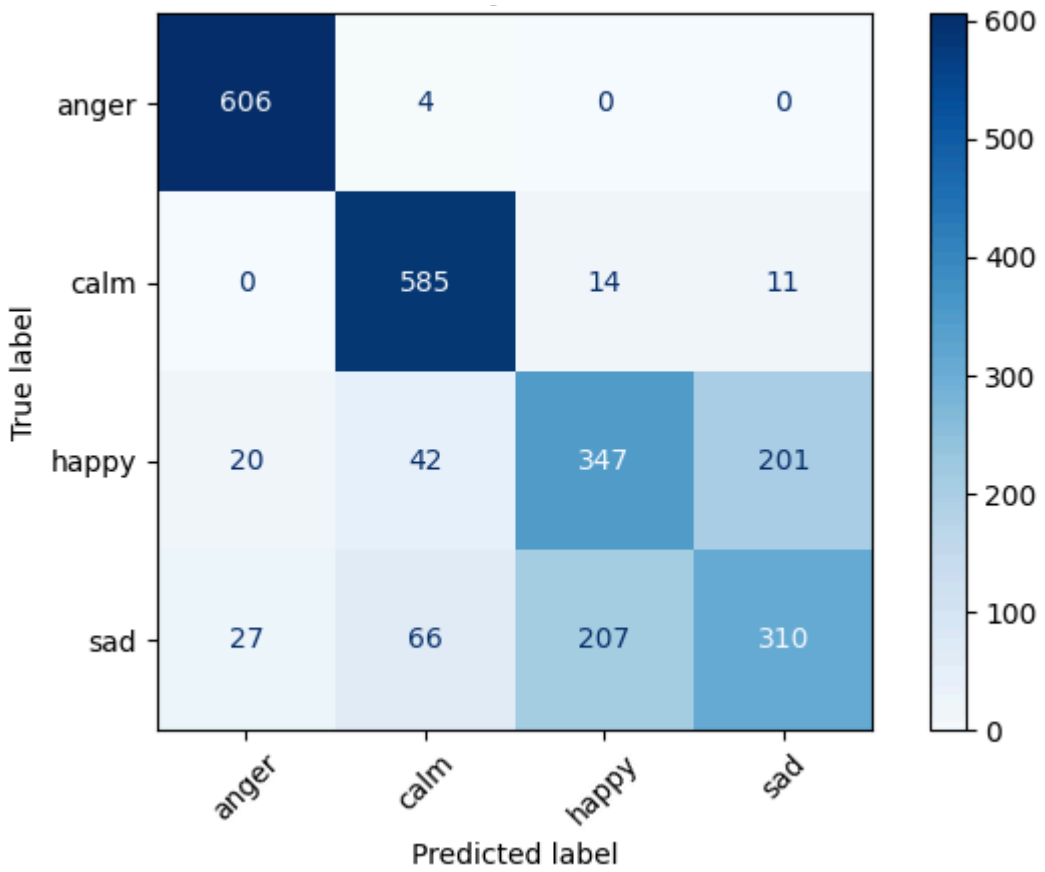
Performance Summary:

Metric	**Value**
Accuracy	0.7574
Exact Match (EM)	0.7574
Macro F1-Score	0.7479
Macro AUC (OvR)	0.9246
Top-2 Accuracy	0.9303
Top-3 Accuracy	0.9902

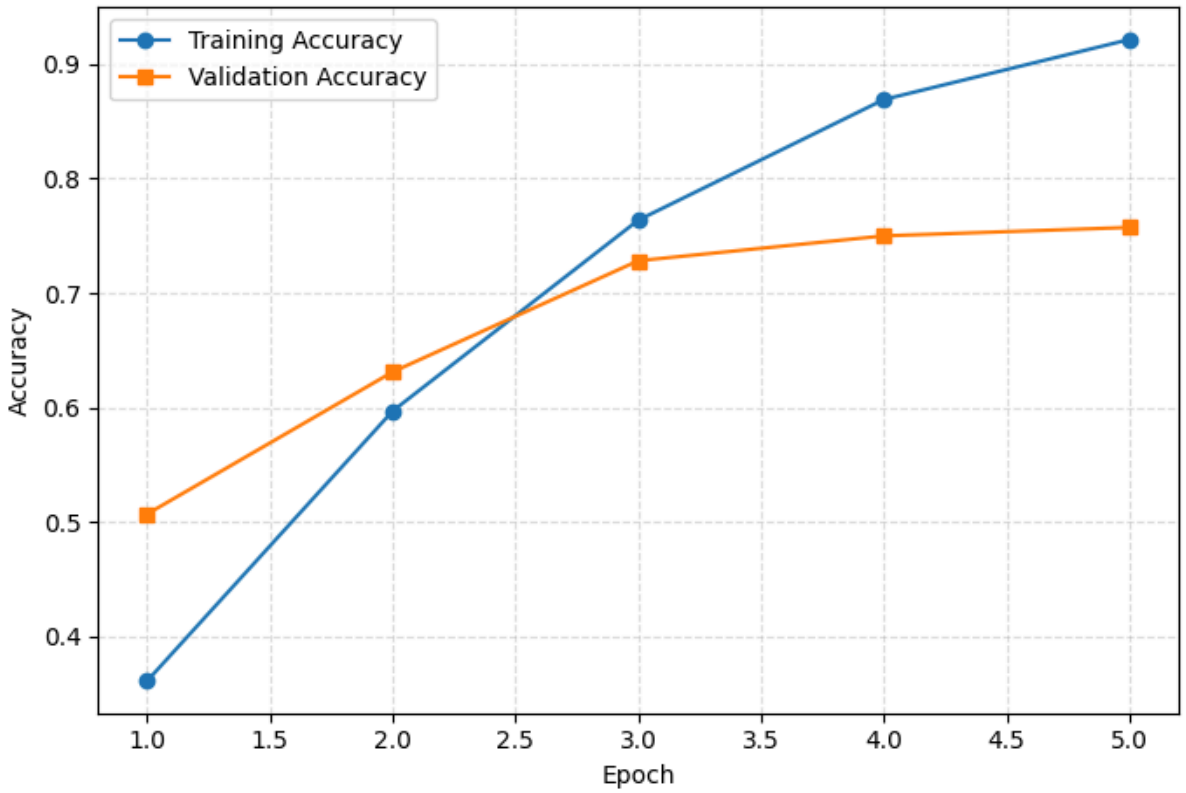
Classification Report:

Class	**Precision**	**Recall**	**F1-Score**	**Support**
anger	0.93	0.99	0.96	610
calm	0.84	0.96	0.90	610
happy	0.61	0.57	0.59	610
sad	0.59	0.51	0.55	610
accuracy			0.76	2440
macro avg	0.74	0.76	0.75	2440
weighted avg	0.74	0.76	0.75	2440

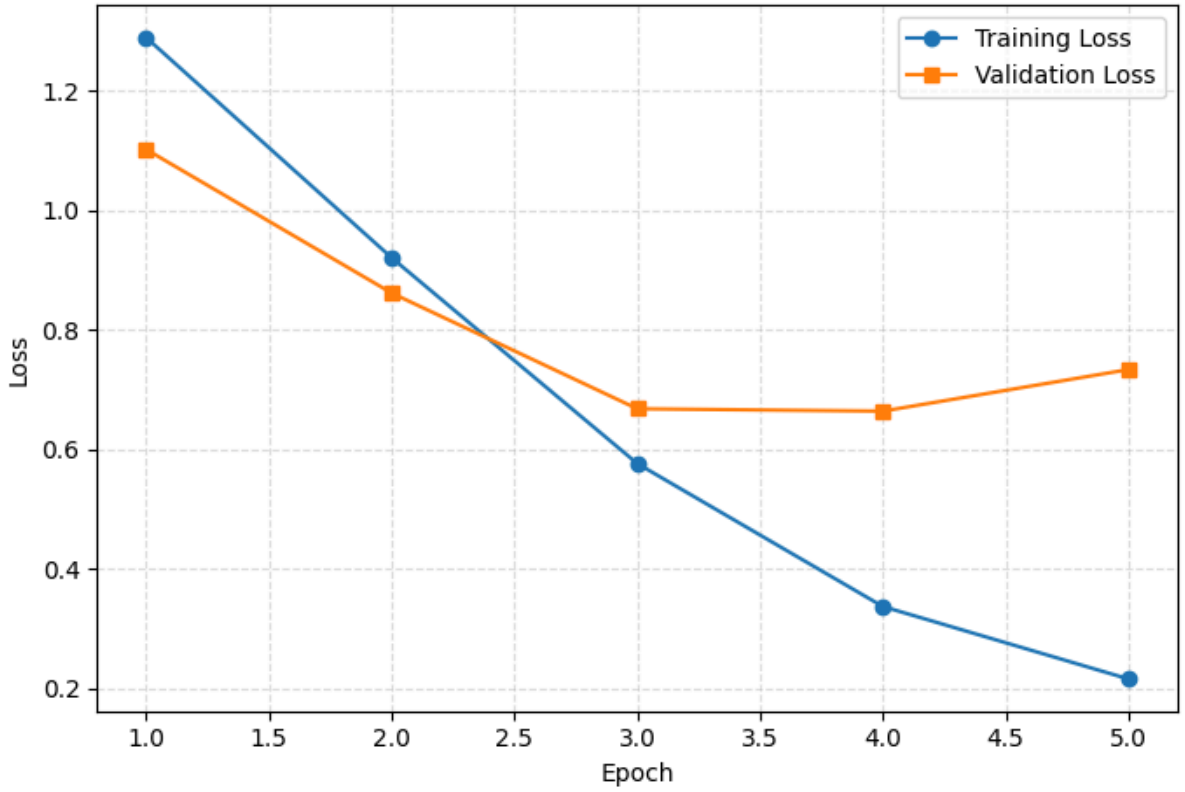
Confusion Matrix:



Training vs Validation Accuracy Curve:



Training vs Validation Loss Curve:



5.2 Results on GoEmotions (6-Class Dataset)

To evaluate whether the performance limitations observed on lyrics were model-based or dataset-based, the same models were tested on the GoEmotions dataset. All models showed significant performance improvements, confirming that dataset characteristics heavily influenced results.

5.2.1 Classical Machine Learning Models (GoEmotions)

Classical approaches performed strongly on the GoEmotions dataset, which contains short, literal, conversational text aligned with TF-IDF assumptions.

Model	Accuracy	Macro F1
Soft Voting Ensemble (LR + SVM + XGB)	0.8870	0.8406
XGBoost	0.8810	0.8386
Linear SVM	0.8805	0.8285
Logistic Regression	0.8765	0.8330

Key Observation: Compared to lyrics (~80%), classical models improved by roughly 8–10 percentage points, confirming the impact of dataset clarity and structure.

The soft voting ensemble on GoEmotions achieved around 88% accuracy, making it a very strong classical baseline on conversational text. It captures the main emotional patterns in the dataset reliably, with robust performance across frequent classes like joy and sadness and reasonable behavior on less frequent ones. Overall, it shows that when emotional cues are clear and literal, traditional TF-IDF features combined with multiple classifiers can generalize very well.

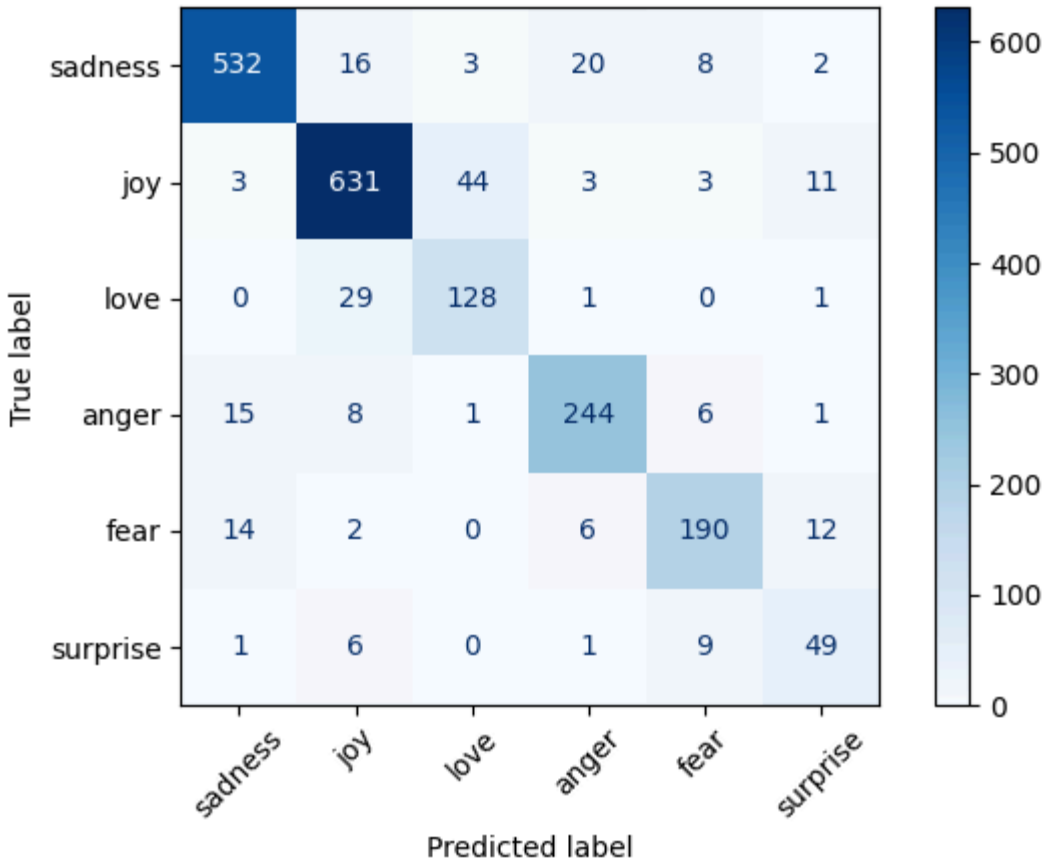
Performance Summary:

Metric	**Value**
Accuracy	0.8870
Exact Match (EM)	0.8870
Macro F1-Score	0.8406
Macro AUC (OvR)	0.9897
Top-2 Accuracy	0.9875
Top-3 Accuracy	0.9960

Classification Report:

Class	**Precision**	**Recall**	**F1-Score**	**Support**
sadness	0.94	0.92	0.93	581
joy	0.91	0.91	0.91	695
love	0.73	0.81	0.76	159
anger	0.89	0.89	0.89	275
fear	0.88	0.85	0.86	224
surprise	0.64	0.74	0.69	66
accuracy			0.89	2000
macro avg	0.83	0.85	0.84	2000
weighted avg	0.89	0.89	0.89	2000

Confusion Matrix:



5.2.2 Hybrid Model

The hybrid model benefited substantially from high-quality, literal language and clearer emotional cues.

Embedding Backbone	Classifier	Accuracy	Macro F1
DistilRoBERTa (j-hartmann series)	XGBoost	0.8800	0.8233

Key Observation: Hybrid model matched classical models in performance and showed strong stability, confirming the benefit of clean semantic embeddings on high-quality datasets.

The hybrid model, using emotion-tuned DistilRoBERTa embeddings with an XGBoost classifier, reached a similar accuracy of about 88% on GoEmotions. Compared to pure classical models, it benefits from richer semantic representations, which help it handle nuanced phrases and context-dependent emotions more gracefully. Its stability and competitive performance suggest that, on clean conversational data, combining high-quality embeddings with a strong tree-based classifier is an effective alternative to both plain TF-IDF and full LLM fine-tuning.

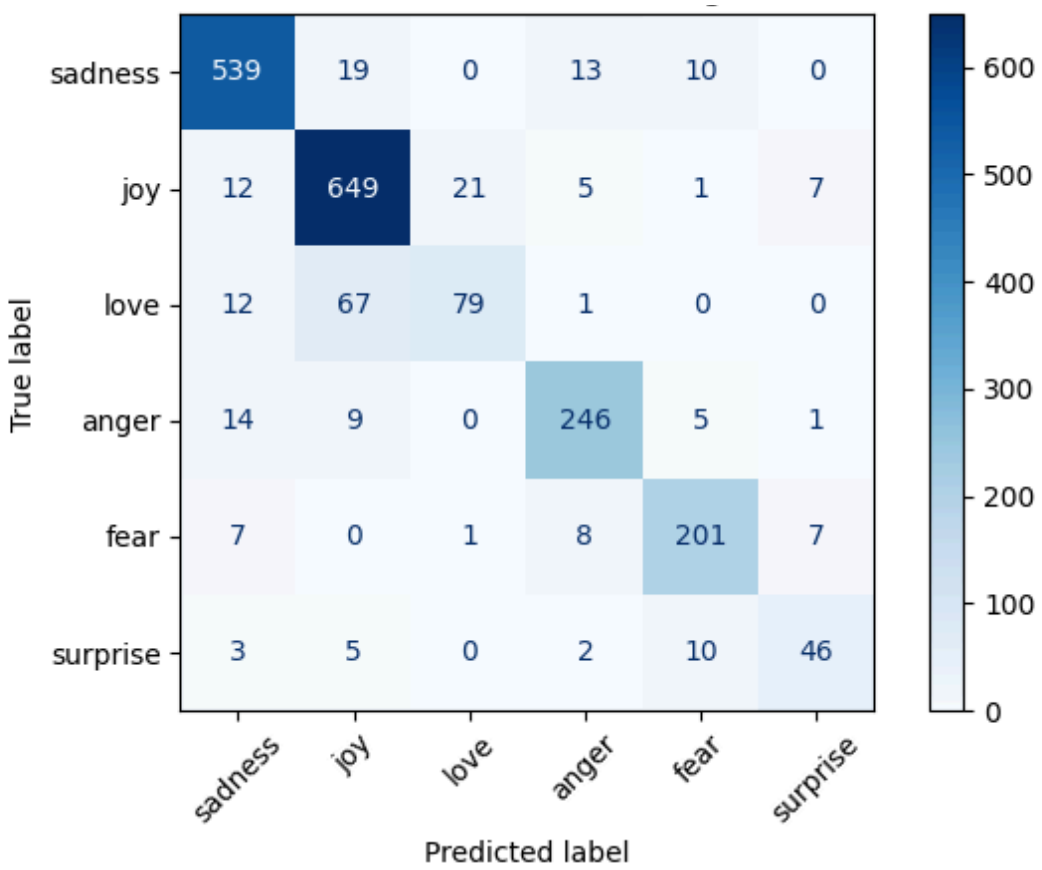
Performance Summary:

Metric	Value
Accuracy	0.8800
Exact Match (EM)	0.8800
Macro F1-Score	0.8233
Macro AUC (OvR)	0.9842
Top-2 Accuracy	0.9770
Top-3 Accuracy	0.9900

Classification Report:

Class	Precision	Recall	F1-Score	Support
sadness	0.92	0.93	0.92	581
joy	0.87	0.93	0.90	695
love	0.78	0.50	0.61	159
anger	0.89	0.89	0.89	275
fear	0.89	0.90	0.89	224
surprise	0.75	0.70	0.72	66
accuracy			0.82	2000
macro avg	0.85	0.81	0.82	2000
weighted avg	0.88	0.88	0.88	2000

Confusion Matrix:



5.2.3 Transformer-Based LLM Fine-Tuning

The fine-tuned transformer model achieved the highest performance overall, reflecting their inherent capability when trained on text that structurally resembles their pretraining corpora.

Model	**Accuracy**	**Macro F1**
RoBERTa-large	0.9300	0.8946

Key Observation: The model produced the strongest test-set metrics across all model categories.

The fine-tuned RoBERTa-large model achieved the best overall performance on GoEmotions, with accuracy close to 93% and a high macro F1-score. It not only models dominant emotions accurately but also handles minority classes such as love and surprise much better than other approaches, reflecting its ability to leverage subtle lexical and contextual cues. This result highlights how, on a well-structured conversational dataset aligned with its pretraining distribution, a large transformer can surpass both classical and hybrid methods by a clear margin.

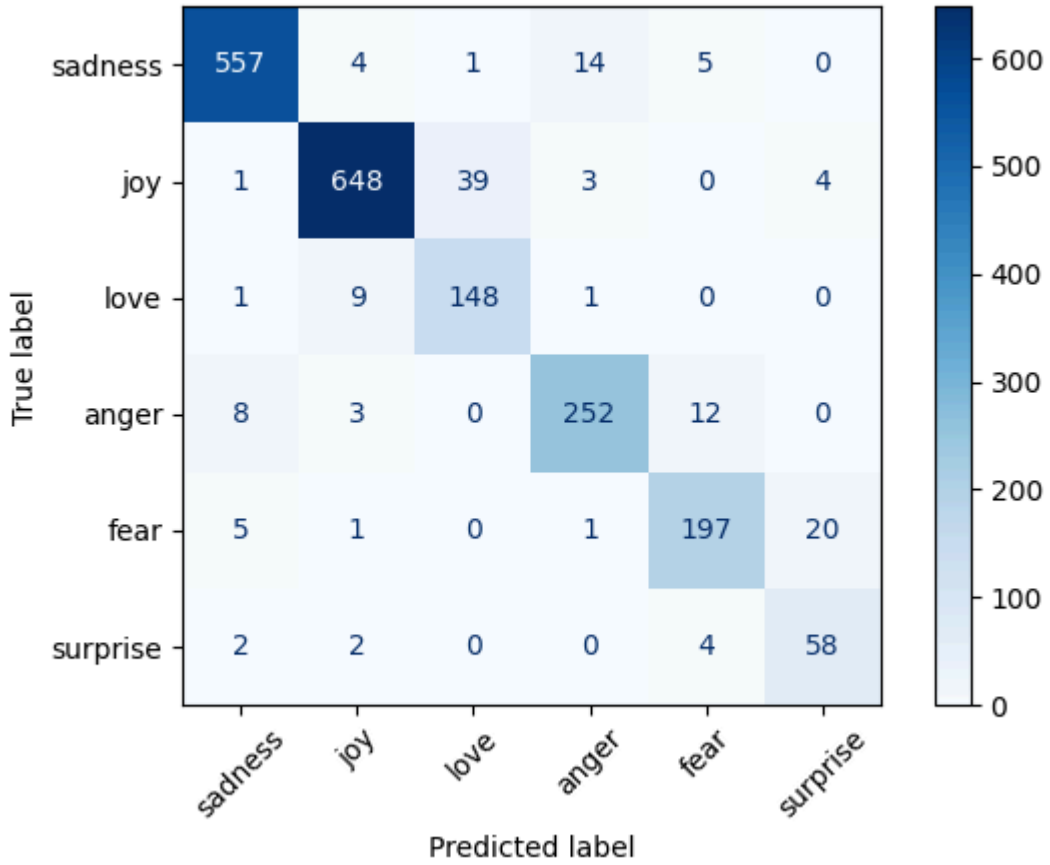
Performance Summary:

Metric	Value
Accuracy	0.9300
Exact Match (EM)	0.9300
Macro F1-Score	0.8946
Macro AUC (OvR)	0.9964
Top-2 Accuracy	0.9965
Top-3 Accuracy	0.9985

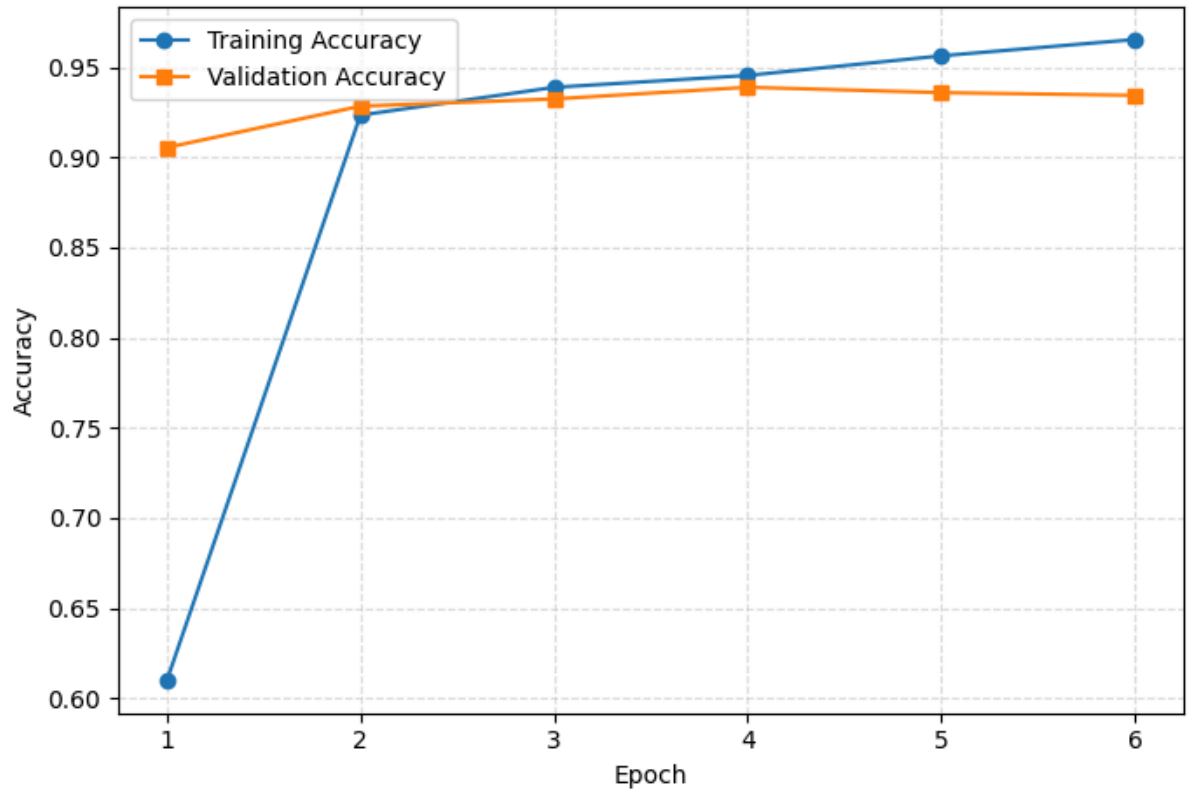
Classification Report:

Class	Precision	Recall	F1-Score	Support
sadness	0.97	0.96	0.96	581
joy	0.97	0.93	0.95	695
love	0.79	0.93	0.85	159
anger	0.93	0.92	0.92	275
fear	0.90	0.88	0.89	224
surprise	0.71	0.88	0.78	66
accuracy			0.93	2000
macro avg	0.88	0.92	0.89	2000
weighted avg	0.93	0.93	0.93	2000

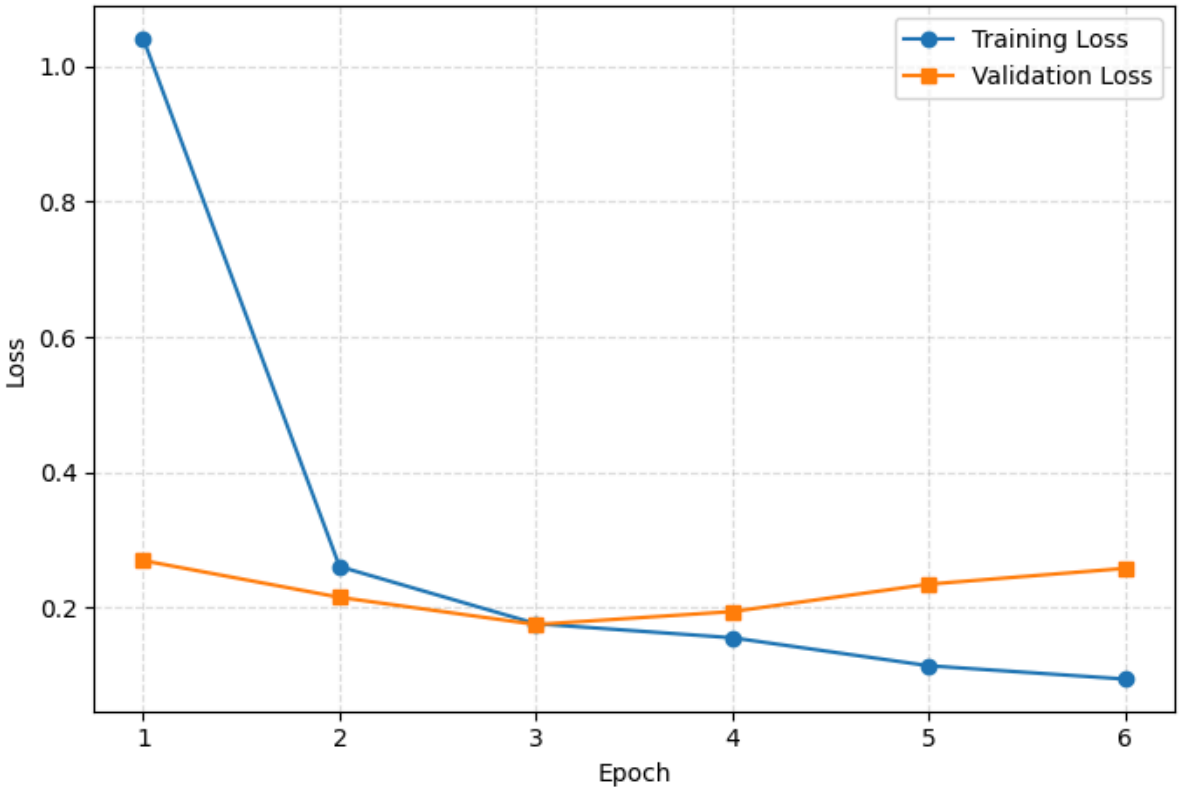
Confusion Matrix:



Training vs Validation Accuracy Curve:



Training vs Validation Loss Curve:



6. Discussion

The findings from this study highlight several important insights regarding the behaviour of classical machine learning models, hybrid embedding-based models and transformer-based LLMs when applied to the task of emotion classification. The results demonstrate that model performance is shaped far more by dataset characteristics than by algorithmic sophistication. This section interprets the results, explains the observed trends and discusses broader implications for lyric-based NLP tasks.

6.1 Impact of Dataset Linguistic Structure

A central observation across experiments is that the lyrics dataset imposed a strict performance ceiling near ~80%, regardless of model complexity. This phenomenon is directly tied to the linguistic properties of song lyrics:

- Lyrics frequently employ metaphor, symbolism and artistic ambiguity, reducing the reliability of literal semantic cues.
- Emotional expression is often multi-layered, making class boundaries inherently fuzzy.
- Lyrics contain non-standard syntax, rhyme patterns and repeated phrases, which differ significantly from the text corpora used to pretrain LLMs.

These properties made the emotional signal within the dataset weaker and noisier, limiting all models' ability to accurately separate emotion classes. As a result, even large-scale transformer architectures converged at performance levels below classical models.

6.2 Why Classical Models Outperformed LLMs on Lyrics

Although transformers are typically expected to outperform traditional models in NLP tasks, this study revealed the opposite pattern on lyrical data. The underlying reasons include:

- TF-IDF captures literal word distributions, which often correlate with emotional tone in lyrics (e.g., "cry," "alone," "rage," "happy"). When emotions are expressed through surface-level word choices, classical models excel.
- Classical models such as Logistic Regression and Linear SVM are less sensitive to noise and do not rely on deep contextual meaning. This makes them more robust to figurative or creative phrasing.
- LLMs rely heavily on contextual semantics learned from prose and dialogue, not poetry. This mismatch degraded their ability to interpret metaphorical emotional cues.

Thus, despite being technologically advanced, LLMs were disadvantaged by domain mismatch, while classical models benefited from the direct mapping between lexical patterns and emotion labels.

6.3 Why Hybrid Models Did Not Surpass Classical Models on Lyrics

Hybrid models were expected to provide a middle ground between classical simplicity and LLM semantic richness. However, on the lyrics dataset, hybrid performance remained slightly below classical models.

The main reason lies in the nature of embeddings:

- Transformer embeddings capture deep semantic relationships, but these relationships are less effective when the underlying text is non-literal or symbolic, as in song lyrics.
- Embeddings compress the meaning of sentences into dense vectors, but when emotional meaning is diffuse or ambiguous, the compressed representation becomes less discriminative.
- Classical classifiers applied over embeddings were unable to recover fine-grained emotional signals from these compressed features.

Therefore, hybrid approaches did not offer meaningful advantages for this domain.

6.4 Cross-Dataset Validation: Evidence of Dataset-Driven Limitations

When the same models were evaluated on the GoEmotions dataset, all performance values rose sharply:

- Classical models improved from ~80% to ~88%
- Hybrid models improved from ~79% to ~88%
- LLMs improved dramatically from ~75% to ~93%

These improvements confirm that the primary bottleneck in the lyrics experiments was the dataset itself, not the models.

GoEmotions consists of short conversational text that is direct, literal and easy to interpret. The emotional cues within the dataset are expressed clearly and with sufficient contextual information, allowing models to distinguish between categories reliably. This aligns much more closely with the training data of modern transformers. Consequently:

- Classical models benefited from clean lexical cues
- Hybrid models leveraged semantically rich embeddings effectively
- LLMs demonstrated their full capabilities, achieving state-of-the-art performance

This cross-dataset comparison strengthens the validity of the project's findings and supports the conclusion that dataset quality and linguistic style are decisive factors in emotion classification performance.

6.5 Implications for Real-World Emotion Classification

The results suggest practical guidelines for selecting model families based on data characteristics:

When to use Classical Models:

- Text is short, noisy, ambiguous, or metaphorical
- Meaning is expressed through overt word choice rather than context
- Dataset size is limited

When to use Hybrid Models:

- Sentences are structurally consistent
- Emotional cues are explicit
- Contextual semantics provide additional clarity

When to use Fine-tuned LLMs:

- Dataset resembles natural conversational language
- Emotional nuance depends on context rather than keyword frequency
- Sufficient computational resources are available

6.6 Limitations of the Study

While the project provides strong empirical evidence, the findings must be interpreted with certain limitations in mind:

- The lyrics dataset contains annotation subjectivity, which may introduce label noise.
- LLM fine-tuning was limited to feasible computational budgets and larger-scale training may yield marginally improved results.
- Only English datasets were evaluated; multilingual lyrics may behave differently.

7. Conclusion

The findings of this study highlight the strong influence of dataset characteristics on the performance of emotion classification models. On metaphorical and stylistically complex song lyrics, all models, regardless of their sophistication, were constrained by the ambiguity and non-literal nature of the text. Classical models demonstrated the most stable behaviour and achieved the highest accuracy within this domain, while hybrid and transformer-based models were unable to fully leverage their semantic capacities due to the artistic structure of the data.

In contrast, all models showed marked immediate improvement on the GoEmotions dataset, where emotional cues were direct, literal and aligned with the linguistic patterns found in large-scale pretraining corpora. The substantial jump in accuracy across classical, hybrid and LLM models reinforces the conclusion that dataset quality and linguistic clarity play a more decisive role than model complexity in emotion classification tasks.

Overall, the study demonstrates that successful emotion classification requires alignment between model architecture and data domain. The experiments underscore the necessity of evaluating NLP systems across multiple datasets to ensure robust, reliable conclusions in research and applied settings.

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